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Investigative practice as a professional project

Interpreting the professional project

The previous chapters have provided abundant evidence for the fact that the historical development of psychological research practice did not proceed along a single track. At the very least, there were two different lines of development. By the 1920s the Wundtian style of experimentation, with its roots deep in the philosophical and scientific traditions of the nineteenth-century German university, seemed to constitute a deteriorating research program within American psychology. As it was fast losing its appeal for all but a few practitioners, the Galtonian research program, strongly linked to practical rather than academic concerns from the beginning, was moving from strength to strength. Not only was it extending its appeal with every passing year, but, as we have seen, it proved itself capable of generating exciting methodological innovations that promised to extend the scope of scientific psychology far beyond what had hitherto been thought possible.

We have also seen that the galloping success of the Galtonian program was in no small measure due to its very direct link to the demands of a significant extradisciplinary market for its products. Nevertheless, it would be a mistake to see the development of a certain style of investigative practice as an essentially passive adaptation to external social requirements. That might have been true of certain individuals, but it was not true of the response of the discipline as a whole. Even on an individual level most practitioners recognized that practical successes would be short lived if they threatened to undermine the prestige of the disciplinary enterprise they represented. If the practice of mental testing, for example, proved to be transparently unreliable, it would do the practitioners no good in the long run, no matter how spectacular the immediate yield might appear to be. The practical effectiveness of psychologists depended quite crucially on

their ability to maintain and enhance whatever standing as experts they had managed to achieve. That standing, in turn, depended almost entirely on their ability to convince their publics that they represented the sacred spirit of Science. Legitimizing their practices in terms of supposedly universal and eternal principles of scientific method was therefore a permanent preoccupation – often to the point of obsession – for all farsighted practitioners.

What exactly was the nature of the product that American psychologists were beginning to market successfully from about 1910? Were their mental-tests goods like material commodities that lost their intrinsic link to the producer once they were sold? Obviously not, for attempts to treat tests in this way led to justified charges of their “misuse” at the hands of “untrained” personnel. Clearly, what the psychologists were marketing were their own skills and their own special competence of which the test materials were merely the external mark. The product that their society valued was *professional expertise*. However, this was a pretty abstract category, and any specific instance of it still had to be established as being the genuine article and not some sort of counterfeit quackery. In other words, the social acceptance of their expertise required that psychologists legitimate their claims in terms of certain widely accepted criteria, which were based on prevailing conceptions of scientific method. It was not enough that their methods yield results that were useful to socially dominant groups; the methods themselves had to be seen to be rational, which, in that particular historical context, meant that they had to bear the hallmarks of science.

In developing their investigative practices psychologists were not only producing instruments for the solution of technical problems, they were also establishing and improving their own status as professionals and as scientists. In a society that places such high value on the methodology of science, the practice of anything recognized as belonging in that category will reflect positively on the prestige and social status of the practitioner. Investigative practices are neither neutral nor disembodied. They carry a social value, and they are inseparable from the social standing of those who enact them.

By the early twentieth century the use of certain investigative practices reflected not only on this or that individual but on a group with the social identity of “psychologists.” What the individual investigator was seen to be doing had potential social consequences for others who shared his or her professional identity. Practitioners within the discipline were increasingly tied to each other by their participation in a *professional project* whose fortunes would potentially affect all of them.¹ This project involved matters of social status, of collective social mobility, of competition for scarce resources, of boundary negotiation with neighboring disciplines, of demarcation from quacks and lay practitioners, as well as related concerns. The kinds of investigative practices with which psychologists were associ-

ated were inseparable from their professional project because these practices played a role in virtually every aspect of the project. They served to distinguish psychologists from certain neighboring disciplines and professions with which there was actual or potential competition, and they served to draw a sharp line between experts and laymen.² Above all, however, investigative practices were strongly implicated in that aspect of the professional project concerned with legitimating the scientific credentials of the discipline.

The criteria for being recognized as scientific had relatively little to do with how the established sciences actually achieved their successes. That involved a set of very complex issues, which remain controversial to this day. What was socially important, however, was the widespread acceptance of a set of firm convictions about the nature of science. To be socially effective it was not necessary that these convictions actually reflected the essence of successful scientific practice. In fact, the most popular beliefs in this area were based on external and unanalyzed features of certain practices in the most prestigious parts of science. Such beliefs belonged to the rhetoric of science rather than to its substance.³ They generally involved the elevation of certain unexamined and decontextualized features of scientific procedure to the status of fetishes. Such sacred and unquestioned emblems of scientific status included features like quantification, experimentation, and the search for universal (i.e., ahistorical) laws. A discipline that demonstrated its devotion to such emblems could at least establish a serious claim to be counted among the august ranks of the sciences.

The contribution of investigative practices to the professional project of psychology involved two sets of problems with often diverging implications. On the one hand, there was the need to develop practices whose products would answer to the immediate needs of socially important markets. But on the other hand, there was the need to establish, maintain, and strengthen the claim that what psychologists practiced was indeed to be counted as science. These two requirements could not always be easily reconciled, and so it was inevitable that there was conflict within the discipline with some of its members placing relatively more emphasis on one or another of these directions.⁴ But in the long run the two factions depended on one another, rather like two bickering partners in a basically satisfactory marriage.⁵ For the very term "applied psychology" reflected the myth that what psychologists put to use outside universities was based on a genuine science, much as engineers based themselves on physics.⁶ Without this myth the claims of practitioners to scientific expertise must collapse. But at the same time, the pure scientist in the laboratory could often command much better support for his or her work if it was seen as linked to practical applications by a society for which utilitarian considerations were paramount.⁷

For American psychology the period between the two world wars was

marked by considerable dissension between its “basic” and “applied” wings, but also by determined efforts to find a compromise that would satisfy both parties. These conflicts were quite clearly reflected on the level of research practice. One orientation, represented by a few prominent individuals and by prestigious publication outlets like the *American Journal of Psychology* and the *Journal of Experimental Psychology*, strongly favored experimental practices and shunned the use of mental tests as well as natural situations and populations. The other orientation, strongly represented in research publications like the *Journal of Applied Psychology* and the *Journal of Educational Psychology*, favored mental tests and made only limited use of experimentation.

The two orientations seemed to draw selectively on different criteria for establishing the worth of their investigative practices. Experimentalists were able to draw on the apparent analogy between what they were doing and what physical scientists were doing.⁸ There was no question that engaging in experimentation for the purpose of discovering universally valid natural laws constituted scientific practice. On this kind of criterion much of what the applied psychologists were doing seemed to be of doubtful respectability but the latter group could always point to the relevance of their work for immediate social and institutional problems.

Nevertheless, this obvious divergence should not blind us to a counter-acting convergence that was operating at the same time. One important expression of this convergence is provided by the experimentalists’ gradual switch from single-subject research to the statistically constituted group data that had always been favored in the more “applied” kind of research. This switch was alluded to in chapter 5. Once experimentalists had accepted the need to organize their work around the production of group data, the gap that separated the investigative practices of the two disciplinary subgroups had been significantly narrowed and a basis for a new disciplinary consensus had been established. In the next two sections we will look more closely at the dialectical interplay of converging and diverging tendencies in the development of investigative practices during the crucial interwar period.

Divergence of research styles

In chapter 5 we distinguished between two kinds of collectivities that were employed in the generation of group data. “Natural” groups were defined by culturally given categories, whereas “artificial” groups owed their definition to the investigator’s intervention. Examples of the former were sex and age groups, while the latter were exemplified by groups defined in terms of scores on some psychometric test or by the experimental treatment they had been given. Two general tendencies were noted (Tables 5.3 and 5.4): an overall trend, during the interwar period, for group data of all

types to replace individual data; and a much greater tendency to use group data in the applied literature than in journals of “basic” research.

If editors and contributors to these two sets of journals were selectively emphasizing different aspects of psychology’s professional project, as discussed in the previous section, we would not expect to find the same mix of group data in both sets. Data pertaining to “natural” groups have a manifest social relevance, which is due to their display of knowledge in terms of culturally familiar categories. For example, one could always count on a significant lay interest in information that pertained to differences among children varying by age, background, and sex. The production of such information would be high on the agenda of those who emphasized the role of psychology as a socially useful discipline, and we would expect the “applied” journals to function as their outlet.

Data generated by the use of experimentally constituted groups, however, did not generally exhibit the same degree of manifest social relevance. Psychologists certainly claimed social relevance for some of the statistical data generated in the course of experimentation, but such claims tended to be more controversial and less accessible to lay understanding than knowledge claims based on the more descriptive use of “natural” groups. What the production of experimental data did have in its favor was the analogy with the activity of “real” scientists.

The differences between “basic” and “applied” journals in respect to their selective emphasis on different types of group data are summarized in Table 8.1. Two points should be noted prior to decoding the significance of these figures. First, the percentages entered in the cells of Table 8.1 refer to the ratio of studies based on each kind of group to the total number of empirical studies published in a particular set of journals. Thus, the sum of the percentages in each row is always less than 100, the difference being made up by studies that used no group data or a mixture of group and individual data. Obviously, the sum of the percentages in a row will be higher the greater the proportion of studies that relied purely on group data. However, the patterns in the overall use of group data have already been discussed in conjunction with Table 5.3 and 5.4. What is of interest now are the differences in the relative use of different kinds of group data.

Second, it should be noted that our coding category of “psychometric groups” does not provide an index of the simple use of mental tests for purposes of social selection. By the 1920s it is actually the “natural groups” category that is mostly made up of studies of this type. What the “psychometric groups” category represents are correlational studies in which distributions of scores on different tests or scales are compared with each other – that is, studies in which the categories of research are entirely generated by the research activity itself. Such studies were of primary interest to psychologists themselves and were at one remove from the apparently immediate social relevance of natural-group studies.

Table 8.1. *Use of different types of group data*

Type of group	<i>Journal of Educational Psychology</i>	<i>Journal of Applied Psychology</i>	<i>Journal of Experimental Psychology</i>	<i>American Journal of Psychology</i>
<i>1914–1920</i>				
Natural	25	37	6	0
Psychometric	19	24	9	3
Experimental	31	15	23	23
<i>1924–1926</i>				
Natural	38	35	16	11
Psychometric	29	36	3	5
Experimental	25	11	25	19
<i>1934–1936</i>				
Natural	45	34	12	8
Psychometric	22	39	2	4
Experimental	27	20	48	44

Notes: Number of group data studies based on each type of group as a percentage of the total number of empirical studies for each time period. In the case of a minority of studies that reported results based on more than one type of group, a coding priority rule was used that ordered the categories of groups in reverse order from the order in which they appear in the table. Thus, if any part of the study relied on experimental groups, that would be the code assigned, irrespective of other types of group that might also occur in the same study. If a psychometric group was used, that category took precedence over natural group. See appendix for explanation of the 1914–1920 time period.

It is clear that there are some striking differences between the “applied” and the “basic” research literature. In the first place, the two applied journals, the *Journal of Applied Psychology* and the *Journal of Educational Psychology*, are much more likely to publish studies with results that are directly attributed to “natural” (i.e., real-life) groups. These journals clearly maintain the tradition of a socially relevant science that attempts to provide knowledge pertaining directly to categories of persons defined by culturally accepted criteria. In the case of one of these applied journals, the *Journal of Educational Psychology*, the exclusive reliance on data pertaining to natural groups even increases from a quarter to nearly a half of all the published empirical studies in the course of the period under review.⁹ The attribution of findings to existing social categories is clearly very marked in the applied literature but relatively weak in the journals devoted to basic experimental research.

However, this does *not* mean that the applied literature lags behind the basic literature in regard to the use of artificial groups. In the *Journal of Educational Psychology*, one-half of all the empirical studies involve attribution of findings to artificial groups from the earliest time period on-

ward. The *Journal of Applied Psychology* reaches this level of reliance on artificial groups in the 1920s, but the basic research journals do not reach it until the following decade. By that time the *Journal of Applied Psychology* has reached a level (59 percent) that suggests a continuation of the trend toward even greater reliance on artificial groups in the future. There appears to be an overall trend in three of the four journals for the use of artificial groups to be distinctly more prevalent at the end of the twenty-year period than it was at the beginning. Interestingly, the exception is the *Journal of Educational Psychology*, which starts with the highest incidence of artificial groups but then moves in the direction of increasing the use of natural groups.

These data make it possible to give a more differentiated account of the increasing tendency to use group data that was noted in chapter 5. In the main, this now emerges as a tendency to make more use of artificial groups in the attribution of data. Increasingly, psychological research constitutes its own collective subjects, which are defined by criteria that are different from the criteria used to group individuals in everyday life. The discipline is busily constructing its own set of categories, which its research practice then imposes on the individuals studied. More and more of psychological knowledge comes to refer to a world in which people are grouped by specially constructed categories that may have only a vague and tenuous connection with the categories that are used to distinguish people in ordinary life.

We can regard any set of psychological research data as a product of two sets of knowledge interests. On the one hand, there is an interest in producing the kind of knowledge that appears to be practically useful to certain potential consumers of the knowledge outside the discipline. But on the other hand, there are interests that aim at the advancement of the discipline, both in a cognitive and a social sense. What is desired here is knowledge that primarily will appear to further the cognitive and technical control of the discipline over its subject matter and will improve its status among the sciences.¹⁰

These two sets of interests are not necessarily antagonistic, although they may at times become so. Intradisciplinary interests may in fact be a reflection of broader societal interests, and may simply be projecting a much longer time frame for the achievement of broadly the same goals. But at any one time the two sets of interests are likely to be clearly distinguishable and to entail different *immediate* consequences for research practice. Thus, the mixture of investigative practices to be found in the discipline, or one of its subdisciplines, at a certain time can be regarded as the product of a particular balance between these sets of interests.

In the field of educational psychology the pressures to produce knowledge of immediate relevance to real groups of children appear to have been highly effective. The relatively close integration of psychological prac-

tice with educational institutions meant that the target populations for research were increasingly defined by those institutions. This integration did not occur at the expense of psychologically defined collectivities, however, but at the expense of the last vestiges of traditional psychological research in this area. Educational psychology accommodated its practices quite fundamentally to the requirements of its market, but in doing so it actually advanced the disciplinary project by making psychological categories a factor to be reckoned with in some important real-world institutional contexts.

By contrast, psychologists in the industrial field had to face quick disillusionment, when the expected profitability of mental tests was not immediately apparent.¹¹ They also faced a major threat from untrained practitioners.¹² Although attempts to demonstrate the direct relevance of findings to real groups continued unabated, the investigative practice of applied psychology during this period seems to have been marked by redoubled efforts at improving its technical sophistication. This would not only increase the distance between professionals and nonprofessionals but might also improve the predictive value of instruments on which any future practical success of the area depended.

But these were special subdisciplinary problems. Returning to the overall pattern of disciplinary practices, we still have to take into account certain differences in the use of artificially constituted collectivities for purposes of data attribution. The major distinction here is between psychometrically and experimentally defined groups. In the former, the group of subjects to whom the research findings are attributed is defined by its possession of certain psychological qualities *elicited* by the psychologist's intervention. By contrast, experimental groups are defined by their having undergone a certain pattern of experimental intervention. The difference between the psychometric and the experimental type depends on the nature of the psychologist's intervention, which depends in turn on the way in which that intervention is interpreted by the investigators. In the one case the observed activity of the subjects is considered essentially as an *effect* of the intervention; in the other case it is considered essentially as an *expression* of some quality of the subjects.

Table 8.1 indicates a massive difference between the applied and the basic research journals in the use of investigative groups constituted by psychometric intervention. Such groups occur much more frequently in the former than in the latter journals, and, if anything, this divergence appears to increase somewhat in the course of time. Conversely, there is more use of experimental intervention as the defining characteristic of investigative groups in the basic-research journals. Our choice of journals appears to have provided information that suggests that there were marked differences within the discipline in regard to the uses of group data. For some, such data were useful in the overall assessment of experimental intervention

because it enabled one to iron out individual differences. But for others such data were needed precisely because it was desired to have some basis for placing individuals in relation to the group.

If we distinguish between the research published in the applied journals and that published in the basic journals, it is clear that during the interwar period it was only the latter group that was undergoing something like a revolutionary development. The pattern for applied-research styles had been essentially established by the end of World War I, and in the ensuing years no fundamental changes occurred. This was basically a Galtonian style of research concentrating on the distribution of psychological characteristics in natural or psychometrically constituted populations. Thus, by this time there were two quite divergent styles of psychological research in existence. One worked with data from individual subjects reacting under laboratory conditions, the other with populations surveyed statistically. The one had the weight of tradition and the mystique of the laboratory behind it, the other was buoyed up by apparent practical success and immediate social relevance.

Emergence of a compromise

If the professional project of American psychology contained contradictory elements, this was entirely due to the fact that it had to come to terms with inconsistent social demands. The material well-being of the discipline depended *both* on its ability to deliver results of immediate practical utility *and* on the successful cultivation of its scientific image. A tendency toward a division of labor resulted, with one part of the discipline emphasizing direct social relevance and the other, scientific respectability. But it would be a mistake to see only the divergence on the surface and to ignore the more subtle unifying tendencies that were also emerging.¹³ In fact, the two major directions were not totally distinct but showed an increasing degree of interpenetration. Faced with divergent social demands, it was only to be expected that practitioners would become more and more adept at finding solutions that would reconcile these demands.

Thus, the new generation of educational psychologists coming to prominence in the early years of the twentieth century was able to eclipse the previous efforts of G. Stanley Hall at "child study" partly because they commanded a more sophisticated scientific rhetoric. It was not just practical relevance that guaranteed success, but practical relevance covered by the mantle of science. In this case the science in question was grounded in the statistical biometry of the Galton-Pearson school. It is true that the psychologists who developed mass mental testing on this basis were responding to administrative requirements, but their successful use of the rhetoric of

science enabled them to insert their own categories into the administrative process and so to advance their professional project.

These developments are relatively clear, but during the interwar period there emerged another kind of synthetic practice that affected experimentalists rather than mental testers. In this case it was the impressive practical successes of Galtonianism that began to influence the practice of psychological experimentation.

Laboratory methods were not abandoned in favor of mental testing, but, increasingly, laboratory data were attributed to groups of subjects rather than to individuals. This trend is particularly noticeable at the end of the period. For the two experimental journals the use of group data in a laboratory context doubles between the mid-1920s and the mid-1930s (see Table 8.1). We have previously noted that there was a corresponding decrease in the attribution of experimental results to individuals during this period (chapter 5). In other words, experimentalists were now less inclined than they had been traditionally to make knowledge claims based on the behavior and experience of individuals and more inclined to base these claims on group averages.

This tendency implied a fundamental shift in the function of the psychological experiment. In the traditional approach, going back to Wundt and other nineteenth-century pioneers, the objects of psychological experiments were individual psychological systems and the use of more than one subject constituted a replication of the experiment. In the new approach the experiment was conceived from the beginning as having a group of subjects as its object. This entailed a Galtonian conception of "error" as involving the variability of performance between individuals, whereas in the traditional approach error involved the variability of response by an individual psychological system. The new kind of experiment no longer sought to analyze or characterize the mode of functioning of individual psychological systems directly. Instead, it sought to establish a statistical norm of response from which individual responses were regarded as deviations. In this respect the new kind of experimentation was much closer to mental testing. Where it differed from the latter was in the retention of experimental control and variation of conditions.

The new approach to psychological experimentation was therefore a hybrid of investigative styles that had previously been established in the discipline. For this reason we may refer to it as a neo-Galtonian style.¹⁴ It is undoubtedly experimental, but it employs experimentation in a Galtonian framework.

What factors might have been involved in the development of this new investigative style? As has already been observed, the apparent successes of an applied psychology that was largely based on a Galtonian use of statistics could hardly have failed to make an impression on the more

traditional sections of the discipline. Increasingly, the latter accommodated themselves to the goal of making predictions about the average response of aggregates. It was this kind of knowledge that seemed to lead to a socially relevant psychological science.

The 1920s were a period during which the older ideals of an academic discipline devoted to the painstaking examination of individual mental processes crumbled before the new goal of putting psychological prediction in the service of large-scale social control. By 1921 three-quarters of the psychologists appearing in *American Men of Science* listed applied interests. More fundamentally, the academic science of psychology was becoming thoroughly penetrated by practical categories of social thought.¹⁵ This was only too clear to a few traditionalists who wrote about the new developments in the tone of critics left behind by the tide.¹⁶ Some, like E. G. Boring, attempted to stem the tide,¹⁷ but their efforts only served to emphasize the direction in which the discipline was moving.

Among the contextual factors that favored these developments, two are well known. First, psychology was not alone during this period in showing a tendency for practical technology to usurp the name of science. In physical science the practical expert was frequently to be found in a preeminent role and the goals of scientific research were often equated with the goals of engineering.¹⁸ Given the prestige of physical science, analogous developments in psychology must have seemed highly legitimate. Second, we are dealing with a period during which large funds began to be available for psychological research from a new source, the private foundations. In particular, the Laura Spelman Rockefeller Memorial moved into the area of social research on a big scale, and psychologists could benefit from this largesse provided their work seemed relevant to the clearly defined goals of those in charge of the funds. What the latter were interested in was "knowledge which in the hands of competent technicians may be expected in time to result in substantial social control."¹⁹ To many it seemed clear that if experimental psychology was to contribute to this kind of knowledge it would have to drop its old knowledge goals and devote itself to the discovery of universal "laws of behavior" that might be useful in charting projects for social control.²⁰

What was attractive to experimentalists about the new methodology was that it allowed them to sidestep the problem of individual differences. In the traditional approach such differences were generally handled in three ways. In the first place, there were persistent attempts to refine the control of conditions so that differences among individuals would be minimized or even eliminated. If this failed, one attempted to find reasons for the peculiarities of individual response patterns – special features of the person's visual apparatus or past experience, for example. Finally, there was always a strong tendency in the traditional approach to limit the scope of experimental psychology to the investigation of those phenomena for which

differences among individuals could be successfully minimized or rationally accounted for. This approach meant a very serious restriction on the expansion of the discipline and a refusal to pursue the investigation of the kinds of psychological problems that were relevant to potential lay consumers of the products of psychological research. By subjecting individual differences to statistical treatment as error variance, the new experimental psychology was able to cast aside these restrictions and to make knowledge claims about "laws of behavior" that were supposed to apply across individuals (and also across situations). Aggregate data provided the empirical basis for such "laws," so that they need not be instantiated in the behavior of a single living individual. The locus of operation of abstract psychological laws was an abstract individual who was the product neither of nature nor of society but of statistical construction.

The purpose of psychological experimentation had therefore undergone a change. In the older kind of practice one manipulated experimental conditions in order to test hypotheses about the processes going on in individual psychophysical systems. Now, the direct purpose of experimentation was to make predictions about how certain variations in conditions affected the response of an abstract individual. Because in practice such an individual was statistically and not psychologically real, questions of psychological inference very easily became transformed into questions of statistical inference, a development to which we shall return in the following chapter.

A comparative perspective: The German case

The scope of our historical interpretation has been confined to data derived from a single social context, the American one. But an examination of analogous aspects of the investigative practices of psychology in a different social context during this period is possible because there was a considerable amount of psychological research in Europe as well. However, only in Germany was the sheer volume of work large enough to allow us to perform a meaningful statistical analysis that would be comparable with the analysis we have made of the American research literature. German psychology will therefore provide our comparison case.

As before, the journal literature in different areas of psychology will be examined for its reliance on statistical data based on aggregate data from groups of subjects. The first area to be looked at is that of industrial psychology, the area in which the national differences turn out to be least marked. In sample volumes of the journals *Industrielle Psychotechnik* and *Psychotechnische Zeitschrift* for 1924–1927, the proportion of empirical studies relying predominantly on group data turns out to have been consistently between 60 and 65 percent. Much the same is true (60 percent) for *Industrielle Psychotechnik* in the years 1934–1936. (The other journal

ceased to be a significant outlet for empirical studies.) These figures are lower than the corresponding figures for the *Journal of Applied Psychology*, which exceeded 80 percent (Table 5.4). The difference is probably due to the fact that at this time parts of German industry were particularly interested in the rationalization of production through the use of what was to become known as human engineering. This often involved intensive individual studies rather than group data.²¹ Nevertheless, the administrative context that provided the bulk of the market for industrial psychology appears to have ensured a preference for statistical group data in both the American and the German case.

When we turn to the educational literature the German-American differences become rather more pronounced. In the *Zeitschrift für pädagogische Psychologie* the proportion of empirical studies based predominantly on group data lies at 53 to 54 percent both for the period 1911–1913 and 1924–1926. (The number of empirical studies is too low for meaningful comparison in the 1934–1936 period.) In the *Zeitschrift für angewandte Psychologie*, whose applied empirical studies fell almost entirely into the field of child and educational psychology, the proportion of such studies relying predominantly on group data was 56 percent for the period 1915–1916 and 53 percent for the period 1924–1926. These figures are not only considerably lower than the corresponding figures for the *Journal of Educational Psychology* (75 percent for 1910–1912 and 91 percent for 1924–1926), but they also show no tendency to increase with time.

What kinds of factors might be at work here? One possibility that must be considered is that the appropriate statistical techniques for the analysis of psychological group data were more readily available to American investigators, because the primary source of these techniques was in Great Britain. Such techniques would enhance both the utility and the apparent scientific status of group data, and so might well encourage their use. Now, it is true that by and large the level of statistical sophistication tends to be higher in the American than in the German applied literature of this period. However, this seems to be less a question of the availability of the relevant information than of its reception. The relevant technical information was widely disseminated through such vehicles as well-known textbooks and publications in the specialist journal literature.²² The influence of Galtonian conceptions of human variability is by no means absent in the German research literature of the period, but it never remotely approaches the overwhelming impact it had on the parallel American literature.

In both Germany and the United States the market for the products of educational psychological research was based on an expanding system of universal education that was adapting itself to the requirements of industrial capitalism. In both cases the effects of this market can be detected, not only on the level of psychological content, but also on the level of inves-

tigative practices. These frequently become geared to the production of a kind of psychological knowledge that is relevant to administrative decisions concerning the selection of persons and of programs in terms of the criterion of performance efficiency.

However, the social situation in Germany appears to have imposed far greater limitations on this process than in the American context. In the latter case education had never been an important source of social status as it had been in nineteenth-century Germany, where rigid segmentation among different types of schools was intimately tied to rigid social class distinctions.²³ In Germany powerful and well-established social mechanisms traditionally governed the selection, both of individuals and of programs, within the educational system. Education officials had a vested interest in maintaining these traditional mechanisms and seldom saw any need to replace or even supplement them with the new mechanisms that applied psychology could provide. The situation did not change substantially during the Weimar period, because of the political weakness of the forces of educational reform.²⁴ Although a limited market for the products of applied psychology did exist, it bore no comparison to the vast education industry that was the powerhouse of the rapid growth of American applied psychology. Instead of functioning as a repository of preindustrial patterns, as it did in Germany, American education quickly adapted itself to provide an almost perfect reflection of the requirements of the new industrial order.²⁵ As we saw in chapter 7, the chief agents of this process were the new educational administrators who provided applied psychology with its most important and most reliable market.²⁶

By contrast, the main consumers of educational psychological research in Germany appear at first to have been classroom teachers. The teachers' associations (*Lehrervereine*) of some of the larger cities played an extremely active role in promoting the dissemination of the products of this research. In a few key instances they established research laboratories and subsidized psychological research from their own funds. The first of these was the Institute for Experimental Pedagogy and Psychology of the Leipzig teachers' association, founded in 1906. This was quickly followed by similar initiatives in Berlin and Munich; other important centers were Breslau and Hamburg.²⁷ Here there was a direct collaboration of teachers and academics guided by a common interest in classroom teaching and by the goal of persuading the authorities of the need for educational reform. Unlike administrators, classroom teachers were directly concerned with psychological processes in the minds of individual children and therefore had an interest in psychological research conducted on this basis.²⁸ This is not to say that group data were irrelevant to their practice, but merely to indicate that these interests were tempered by other concerns. Thus one reason for the difference between German and American educational psychology in

the overall pattern of investigative practices may have to be sought in the different professional alliances that were feasible and advantageous in these two cases.

However, one must be careful to avoid the kind of oversimplification that would result from considering the educational system as a purely external market for the discipline of psychology. The key figures in the adaptation of psychological investigative practices to the requirements of this market were after all themselves products and functioning parts of the educational system, by virtue of their academic qualifications and appointments. Thus, certain characteristics of the system were not experienced by these individuals as external demands but as internalized values. Among those German academics who took the lead in advancing the cause of applied psychology this often expressed itself in a characteristic duality of goals. The same men who pioneered the advancement of psychological techniques as instruments of social control, and of quantitative methods for providing administratively useful knowledge, often devoted much of their energies to the pursuit of philosophical goals that had been sanctified by the humanistic traditions of German higher education.²⁹ Such split loyalties contrast rather sharply with the single-minded devotion to the ideal of calculated efficiency and rationalized performance that was so common among their American counterparts.

We are therefore led back from a consideration of market conditions, external to the discipline, to the question of intellectual interests operating within the discipline. These become particularly important when we enlarge the scope of our inquiry to consider the influence of the investigative practices of applied psychology on the discipline as a whole. It was shown earlier that in American psychology the fundamental shift from the analysis of psychological processes in individuals to the analysis of the distribution of psychological characteristics in populations had significantly affected investigative practices in the area of "pure" research by the 1930s. Can an analogous effect be demonstrated in the German literature of the time? The answer is yes – in a very limited way. During the years 1934–1935 the proportion of empirical studies relying entirely on group data reached a mere 13 percent in *Psychologische Forschung*, 20 percent in the *Zeitschrift für Psychologie*, and 23 percent in the *Archiv für die gesamte Psychologie*. Although these figures are far below the corresponding American figures, where this approach already accounted for more than half the published studies, they do represent a small but consistent increase over the German figures a decade earlier. For the period 1924–1926 the proportion of studies relying solely on group data varies from 0 in *Psychologische Forschung* to 13 percent in the *Archiv für die gesamte Psychologie*. This approach to psychological research was always less heavily represented in the German than in the American research literature. However, a slight tendency to

follow the lead of applied psychology in adopting this approach does eventually become noticeable.

The difference in the historical development of investigative practices in the American and the German case is not an absolute one. Analogous forces seem to be at work, but their relative weight is very different. In both cases the research literature shows a mixture of fundamentally different research practices, which are geared to the production of different kinds of psychological knowledge. In what I have previously called the Wundtian model, psychological research aims to analyze processes located in individuals, whereas in the Galtonian model the concern is with the distribution of psychological characteristics in populations, and the placing of individuals with respect to such group characteristics. Both in American and in German psychology the Galtonian approach is strongly favored by the emergence of an important market for the products of psychological research. Also, in both cases there is a detectable shift in the general research literature toward the Galtonian model.

However, these effects appear to be much stronger in the American than in the German case. Two kinds of factors are likely to account for this difference. In the first place, certain differences in the social role and structure of the educational system had the effect of turning American education into a far more powerful market for psychological products and services than was the case for German education, at least during the first half of the century. But this does not account for the curiously ambivalent response on the part of German academics interested in applied psychology, nor does it account for the extremely weak effect that the Galtonian model of psychological research had on the discipline as a whole. For an explanation of these phenomena we have to consider the way in which members of the discipline defined their professional project. Among American psychologists, there was an early recognition of the fact that the progress of their discipline depended on its ability to supply knowledge that would be useful for the practical management of human affairs. Such early pioneers as G. Stanley Hall and James McKeen Cattell already grasped that the pursuit of this goal would require a switch to research practices that yielded statistical information about defined groups of individuals. Thus, the response to the requirements of the educational market was rapid and unambiguous. This did not mean that there was unanimity within the discipline, but simply that there existed a convergence of external and internal factors that proved decisive for the overall development of the discipline in the long run.

By contrast, German academic psychologists found themselves in a situation where their enterprise was obliged to legitimate itself, in terms of the values represented by a well-entrenched traditional intellectual elite, if it was to prosper.³⁰ On the level of research practices there was no chance

of accomplishing this by means of methods that were seen as a kind of superficial census taking that negated human individuality as well as the traditionally recognized mandate of psychology to explore the individual mind and personality.³¹ The weight of such objections would have been all the greater, as the institutional separation of psychology and philosophy had not become the norm at German universities. Academic psychologists often held appointments in philosophy, and in many cases their own intellectual commitments matched these institutional arrangements. Whether their philosophical position owed more to the humanistic tradition or to the newer irrationalist currents did not matter; in either case their distaste for a numerizing approach to human individuality was predictable.

As in the United States, there were pulls on the emerging discipline from a very different direction, that of psychotechnology and its various areas of application. Here, philosophical scruples counted for nothing, and the only tribunal to which statistical techniques had to answer was one guided by norms of pragmatic effectiveness. The question now became one of which set of interests would set its stamp on the discipline as a whole. Both in Germany and in the United States academic psychologists played the crucial role in determining norms of good disciplinary practice. The difference lay in the ties, institutional and economic, that academic psychologists had with potential centers of influence.³² In Germany the separation between "pure" science and technology, although not as strong as it had been in the nineteenth century, still went very deep, both institutionally and subjectively. Technical studies were segregated in the *Technische Hochschulen*, or Technical Universities, at some of which applied psychology did find a home. However, during the Weimar period at least, the intellectual leaders of the discipline were not to be found there but at the regular universities, where they were immune to any influences from the dirty world of industry. The barriers, both institutional and subjective, against any adoption of ideas or practices from that quarter were formidable. A very different situation existed in the United States, where institutional pressures from a number of sources – university governments; research foundations, both public and private; bureaucracies – all converged to support the orientation of investigative practice toward managerial concerns.

Although it is convenient, for the sake of conceptual clarity, to make analytic distinctions between factors external and factors internal to the discipline, the limitations of such distinctions must be clearly recognized. Internal definitions of disciplinary goals are themselves adaptations to external conditions, and conversely, disciplinary projects contribute to the formation of external markets for their products. From the viewpoint of the process of professionalization, cognitive schemata and technical practices can be treated as a resource,³³ whereas in the present analysis it is precisely these "resources" that are regarded as problematic. The con-

stellation of intellectual interests that constitutes the professional project faces both outward and inward. These interests mediate between the possibilities and demands that confront the community of practitioners and the range of investigative practices accepted as legitimate within the community. These practices in turn circumscribe the kind of knowledge that the discipline is able to produce and define the limits of that knowledge.

9

From quantification to methodolatry

Twentieth-century psychology's infatuation with numbers would not have blossomed into an all-consuming love affair had it not been for the usefulness of statistical constructions based on group data. Statistical constructions represent work done on some raw material that already has a numerical form. However, that form involves more than the counting of heads. The individuals who are counted must also be endowed with countable attributes. Where the attributes are physical, like height or weight, their transformation into numerical form depends on social practices developed long ago, but where one has to deal with psychological attributes the corresponding measuring practices represent a relatively recent development. This development was initially independent of the special use of statistical information that has occupied our attention so far. It is therefore necessary to step backward in time to consider the nature of the practices that made it possible to impose a numerical form on psychological attributes.

What is the difference between one and two?

Because it is rare for human experience and activity to take on a quasi-numerical form spontaneously, psychologists who insist on working with such forms must impose them on their subject matter. One way of doing this is to apply the numerical net post hoc, after the psychological event has taken place and has produced some record. The number of word associations with a particular thematic content can be counted, for instance, either for a particular individual or across individuals. This kind of quantification was attempted by Galton and practiced on a large scale by G. Stanley Hall.¹ Here the production of numerical data depended on the researcher establishing certain thematic categories and then performing a